

TEST PLAN

Project Name

API, Database & Automation Testing of Student Management System

1. Objective

This project's goal is to assess the Student Management System's functionality, dependability, and data integrity through automated testing, database validation, and API testing. The purpose of the testing is to make sure that the system retains accurate and consistent data and that all major functions function as intended.

2. Project Description

Exam scores, attendance, course registration, and student data are all managed by the Student Management System. Testing involves automating important user workflows, confirming database entries, and validating APIs.

3. Scope of Testing

In Scope

- User Registration
- User Login
- Student CRUD Operations
- Course Management
- Attendance Management
- Result Management
- API Validation
- Database Validation
- Automation Testing

Out of Scope

- Performance Testing
- Security Testing
- Load Testing
- Mobile Application Testing

4. Testing Types

- Functional Testing
- API Testing
- Database Testing
- Automation Testing
- Smoke Testing
- Regression Testing
- Sanity Testing

5. Test Environment

ITEM	DETAILS
API Tool	Postman
Database	MySQL
Automation Tool	Selenium WebDriver
Browser	Google Chrome
Operating System	Windows 11

6. Modules Covered

User Module

- Registration
- Login
- Logout

Student Module

- Add Student
- Update Student
- Delete Student
- View Student

Course Module

- Add Course
- Update Course
- Delete Course

Attendance Module

- Mark Attendance
- Update Attendance
- View Attendance

Result Module

- Add Result
- Update Result
- View Result

7. Entry Criteria

- APIs are available.
- Database is accessible.
- Test data is available.
- Required tools are installed.

8. Exit Criteria

- Planned test cases are executed.
- Defects are documented.
- Reports are prepared.

9. Deliverables

- Test Plan
- Test Cases
- Postman Collection
- SQL Validation Scripts
- Automation Scripts
- Execution Report
- Bug Report
- README

10. Risks

- API downtime
- Database connectivity issues
- Invalid test data
- Browser compatibility issues

11. Conclusion

This test plan describes how to validate the Student Management System using automated, database, and API testing. The functioning and data integrity of the program will be ensured if the planned tasks are successfully completed.